Selected Papers

Theoretical Study on the Excited Electronic States of Coronene and Its π-Extended Molecules Using the Symmetry-Adapted Cluster-Configuration Interaction Method

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The excited electronic states and optical absorption spectra of coronene ($C_{24}H_{12}$), hexa-*peri*-hexabenzocoronene (HBC) ($C_{42}H_{18}$), and circumcoronene ($C_{54}H_{18}$) were studied using the symmetry-adapted cluster-configuration interaction (SAC-CI) method. For coronene and HBC, the SAC-CI calculations reproduced the experimental spectra well and predicted optically forbidden excited states. For HBC, the symmetry lowering enhanced the intensity of the S_2 state that corresponds to the *p*-band, and the SAC-CI calculation predicted the existence of the second and third optically allowed states around the β-band region near 4.0 eV. For circumcoronene, the SAC-CI calculation predicted a strong absorption of the β-band in the visible light region. The mechanisms of energy splitting for the HOMO–LUMO transition were investigated. Electron correlation was the most important factor for the energy splitting between the lowest and the next-lowest states. Configuration interaction with single excitations (CIS) calculations could not correctly predict the relative energies of these states in coronene and circumcoronene. For HBC, on the other hand, the CIS calculation provided the same energy order as the SAC-CI calculation.

The electronic and optical properties of polycyclic aromatic hydrocarbons (PAHs) have received increased attention because of their scientific and practical importance.[1] Large PAHs can be regarded as models of graphene whose unusual physical properties are undoubtedly remarkable.[2–4] Various applications of PAHs are being intensively studied as functional molecular materials.[5,6] The optical absorptions of PAHs are also investigated in astrophysics because PAHs could be present in the interstellar medium.[7,8] In this paper, we report studies on the excited electronic states and optical absorptions of coronene ($C_{24}H_{12}$), hexa-*peri*-hexabenzocoronene (HBC) ($C_{42}H_{18}$), and circumcoronene (*cir*-coronene) ($C_{54}H_{18}$); their molecular structures are shown in Figure 1.

Coronene is the most symmetric extension of benzene, comprising six *peri*-fused benzene rings. Its optical properties have been studied by experimental and theoretical methods.[9–13] In the same manner as for other PAHs, low-lying ultraviolet–visible (UV–vis) absorptions of coronene are characterized by three types of valence π–π^* transitions: α-, *p*-, and β-bands.[1] They correspond to electron transitions between two highest occupied molecular orbitals (HOMOs) and two lowest unoccupied molecular orbitals (LUMOs). In terms of group theory, the character of these transitions is the same as that of the valence π–π^* transitions in benzene.

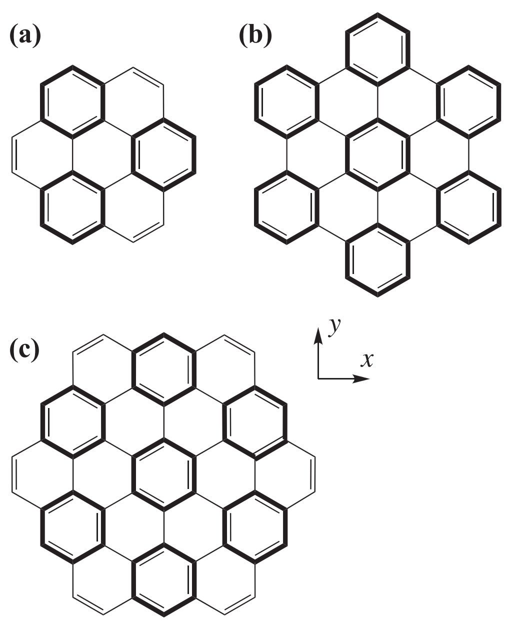

Figure 1. Molecular structures of (a) coronene, (b) HBC, and (c) *cir*-coronene. The bold lines indicate the benzenoid structure of Clar's sextet rule.

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Table 1. Valence Hartree–Fock MOs of Coronene, HBC, and *Cir*-coronene with D95(d) Basis

Orbital	Coronene		HBC		*Cir*-coronene	
	Energy[a)]	Rep[b)]	Energy	Rep	Energy	Rep
HOMO−4	−11.57	a_{2u}	−9.07	e_{1u}	−9.17	e_{1g}
HOMO−3	−9.99	b_{1g}	−9.03	e_{1g}	−8.47	a_{1u}
HOMO−2	−9.53	b_{2g}	−8.90	b_{2g}	−8.13	a_{2u}
HOMO−1	−9.09	e_{1g}	−7.54	a_{1u}	−7.76	e_{2u}
HOMO	−7.08	e_{2u}	−6.80	e_{1g}	−6.15	e_{1g}
LUMO	1.52	e_{1g}	1.20	e_{2u}	0.42	e_{2u}
LUMO+1	3.00	e_{2u}	1.43	b_{1g}	1.84	e_{1g}
LUMO+2	3.51	a_{2u}	2.50	a_{2u}	2.45	b_{2g}
LUMO+3	4.67	a_{1u}	2.71	e_{2u}	2.47	b_{1g}
LUMO+4	5.59	b_{2g}	3.32	e_{1g}	2.66	e_{2u}

a) Orbital energy in eV. b) Orbital symmetry representation.

HBC is a symmetric extension of coronene. According to Clar's sextet rule,[14] resonance stabilizations for HBC and coronene are different. Coronene has only three benzenoid structures. In contrast, HBC has seven-benzenoid structures, and it is categorized as a fully benzenoid PAH[1] (Figure 1). HBCs have been increasingly studied for application to functional molecular materials, mainly because of their availability for preparing various derivatives and their self-assembly character.[6,15–17] *Cir*-coronene is another symmetric extension of coronene comprising the same seven-benzenoid structure as HBC. *Cir*-coronene could be present in hydrocarbon flames;[18] however, experimental research on this molecule has been very limited. Mainly theoretical and computational chemists have taken an interest in this molecule as a model of large planar π-systems.[19–23]

Recently, we have studied excited electronic states of π-extended phthalocyanines[24,25] and C_{60} fullerene[26] using the direct symmetry-adapted cluster-configuration interaction (SAC-CI) method.[27] These studies showed the importance of many-body effects for prediction of many excited states in large conjugated systems. In particular, the energy order of excited states was sensitive to the accuracy of the theories.

In this study, we applied the direct SAC-CI method to coronene and its π-extended systems. The optical absorption spectra were assigned based on the SAC-CI calculations. The effect of symmetry lowering in HBC was considered. The symmetry lowering could explain the relatively high intensity of the *p*-band that could be assigned to the 1^1B_{1u} state. The mechanisms for the energy splitting of the lowest $^1B_{1u}$ and $^1B_{2u}$ states were investigated using frozen orbital analysis (FZOA).[28–30] We demonstrated the importance of the electron correlations for the energy splitting of the $^1B_{1u}$ and $^1B_{2u}$ states. The configuration interaction with single excitations (CIS) calculations could not correctly predict the relative energies of these states in coronene and *cir*-coronene. For HBC, on the other hand, the orbital relaxation was also important, and the CIS provided the same energy order as the SAC-CI calculation.

Computational Details

The equilibrium molecular geometries in the ground state were optimized using the B3LYP/6-311G(2d,p) method[31,32] in the D_{6h} point group. The obtained structures were confirmed to be a local minimum by vibrational analysis except for HBC. As previously reported, the optimized D_{6h} structure of HBC has imaginary values of frequencies and lead to a nonplanar structure with D_{3d} symmetry.[33] For HBC, therefore, both D_{6h} and D_{3d} structures were used for the SAC-CI calculations. The vertical electronic excitation energies and oscillator strengths were calculated by the direct SAC/SAC-CI method[27,34,35] with the D95(d) basis.[36] The 1s electrons of C atoms and their complementary virtual molecular orbitals (MOs) were excluded as the frozen core approximation. The perturbation-selection thresholds were $\lambda_g = 5 \times 10^{-7}$, and $\lambda_e = 5 \times 10^{-8}$ for the ground and excited states, respectively.[37] The nonvariational SAC-CI secular equations were solved by an iterative method with the CIS solutions as the initial vectors. We did not consider the degeneracy of states for the perturbation-selection; therefore, the SAC-CI solutions obtained that should be degenerate may have slightly different energies because of the error involved in the selection procedure. Thus, we averaged the energies of such degenerate states and summed the oscillator strengths. Computations were performed using Gaussian development version.[38]

Results and Discussion

Electronic Excited States. Table 1 shows the valence MOs of coronene, HBC (D_{6h} structure), and *cir*-coronene. For these molecules, the HOMO and LUMO were doubly degenerate. The HOMO to LUMO transition generates four electronic states: $^1B_{1u}$, $^1B_{2u}$, and doubly degenerate $^1E_{1u}$ states. The HOMO–LUMO gap decreased with increasing molecular size. For coronene and *cir*-coronene, the next HOMO (HOMO−1) and the next LUMO (LUMO+1) were also degenerate, while the HOMO−1 and LUMO+1 of HBC were not degenerate. The energy difference between the HOMO and HOMO−1 for HBC was significantly smaller than the differences for coronene and *cir*-coronene. A similar trend was found for the LUMO and LUMO+1. In comparison with coronene and *cir*-coronene, the HOMO and LUMO of HBC were not energetically separated from other orbitals.

The SAC-CI results of coronene are summarized in Table 2. To examine the basis-set dependence, a SAC-CI calculation with triple-zeta basis with one polarization function (TZP) basis set[39] was performed, and the results are also summarized in Table 2. The extension of the basis set to TZP decreased the excitation energies by about 0.1 eV; however, the results with

Table 2. The Excitation Energies (EE), Oscillator Strengths, and Orbital Transitions of Coronene Calculated by the SAC-CI Method

State	D95(d)[a]		TZP[b]	Experiment
	EE/eV[c]	Main transitions	EE/eV	EE[d]/eV (band)
1^1B_{2u}	3.12	H[e] → L[f]	3.03	2.91 (α)
1^1B_{1u}	3.84	H → L	3.77	3.58 (p)
1^1E_{2g}	4.41	H → L+1	4.32	
1^1E_{1u}	4.63 (4.260)	H → L	4.50 (4.094)	4.09 (β)
1^1A_{1g}	4.68	H → L+1	4.57	
1^1B_{2g}	4.70	H → L+1	4.59	
2^1E_{2g}	5.12	H–1 → L, H → L+1, H → L+2	5.01	
2^1B_{2g}	5.21	H–1 → L	5.12	
3^1E_{2g}	5.50	H → L+2, H–1 → L	5.38	

a) SAC-CI/D95(d) results. b) SAC-CI/TZP results. c) Oscillator strength is given in parentheses for optically allowed states. d) 0–0 transition position in THF (Ref. 11). e) HOMO. f) LUMO.

Table 3. The Excitation Energies (EE), Oscillator Strengths, and Orbital Transitions of HBC Calculated by the SAC-CI Method

D_{6h}[a]			D_{3d}[b]		Experiment[c]
State	EE/eV[d]	Main transitions	State	EE/eV	EE/eV (band)
1^1B_{2u}	2.72	H[e] → L[f]	1^1A_{1u}	2.70	2.85 (α)
1^1B_{1u}	3.26	H → L	1^1A_{2u}	3.25 (0.028)	3.35 (p)
1^1E_{2g}	3.41	H → L+1	1^1E_g	3.40	
1^1E_{1u}	3.78 (5.761)	H → L	1^1E_u	3.76 (5.457)	3.62, 3.66, 3.69 (β^0)
2^1E_{2g}	3.96	H → L+1	2^1E_g	3.94	3.83[g] (β^1)
2^1E_{1u}	4.05 (0.014)	H → L+2	2^1E_u	4.05 (0.009)	4.02[g] (β^2)
1^1A_{2g}	4.36	H–1 → L+2, H–2 → L+1	1^1A_{2g}	4.36	
3^1E_{1u}	4.41 (0.024)	H → L+3	3^1E_u	4.41 (0.022)	
2^1B_{1u}	4.46	H → L+3	2^1A_{1u}	4.45	
2^1B_{2u}	4.47	H–1 → L+1, H → L+3	2^1A_{2u}	4.46 (0.006)	
2^1A_{2g}	4.78	H–2 → L+1	2^1A_{2g}	4.78	
3^1E_{2g}	4.95	H–4 → L	3^1E_g	4.96	
4^1E_{1u}	5.09	H–2 → L	4^1E_u	5.08 (0.000)	

a) SAC-CI results with D_{6h} geometry. b) SAC-CI results with D_{3d} geometry. c) Experiment in Ne matrix (Ref. 33). d) Oscillator strength is given in parentheses for optically allowed states. e) HOMO. f) LUMO. g) Assigned to a vibronic band; see text.

D95(d) were parallel to those obtained with TZP. Thus, the overall trends could be reproduced using the D95(d) basis set.

The vibronic structures of the absorption spectrum of coronene have been investigated in detail.[11,40] For the optically forbidden α- and p-bands, the vertical excitations correspond to the position of the 0–0 transition because the 0–0 transitions give the maximum Franck–Condon factor. Consequently, the SAC-CI/D95(d) results reproduced the experimental peak positions of the α- and p-bands with reasonable accuracy. The deviations were about 0.2 eV, and would be mainly attributable to the effect of higher excitations. Indeed, the effects of triple excitations on the excitation energies of coronene were 0.25–0.32 eV calculated with the CR-EOMCCSD(T) (completely renormalized equation-of-motion of coupled cluster with singles and doubles and perturbative triple excitations) method.[41] For the β-band, on the other hand, the deviation between the SAC-CI and experimental values amounted to 0.5 eV. The solvent effect may be significant for this state because of its high excitation energy and large transition dipole moment. Some models show that solvatochromic shifts originated in the dispersion interactions are large in higher excited states with a large transition moment.[42] Appropriate choice of solvation models is essentially important to estimate the solvent effects on these systems by ab initio methods.[43] Such calculations are beyond the purpose of this study. In addition, the SAC-CI with single and double excitations may be also an insufficient approximation for this state. From the SAC-CI calculation, the β-band (1^1E_{1u} state) corresponds to the fourth singlet excited state (S_4). The S_3 state (1^1E_{2g}), which has the character of the HOMO to LUMO+1 transition, exists between the p- and β-bands.

The SAC-CI results for HBC are summarized in Table 3. The excitation energies are almost the same for the D_{6h} and D_{3d} structures. The $^1B_{1u}$ states of D_{6h} symmetry correspond to the $^1A_{2u}$ states of D_{3d} symmetry, and they become optically allowed. For HBC, detailed UV–vis spectra have been measured in rare-gas matrices.[33] The α- and p-bands were observed around 2.9 and 3.3 eV, respectively. Three components of β-band, β^0-, β^1-, and β^2-bands, were observed around 3.5–4.0 eV; each component involves vibronic fine structure.

Table 4. The Excitation Energies (EE), Oscillator Strengths, and Orbital Transitions of *Cir*-coronene Calculated by the SAC-CI Method

State	EE/eV[a)]	Main transitions	Band	TDDFT[b)]	SEMO[c)]
1^1B_{2u}	1.91	H[d)] → L[e)]	α	1.95,[f)] 2.65[g)]	2.19,[h)] 2.13[i)]
1^1B_{1u}	2.43	H → L	p		
1^1E_{1u}	3.01 (7.369)	H → L	β		3.09,[h)] 3.22[i)]
1^1E_{2g}	3.15	H → L+1			
1^1A_{2g}	3.33	H → L+1			
1^1A_{1g}	3.40	H → L+1, H–1 → L			
2^1E_{2g}	3.60	H–1 → L			
2^1A_{2g}	3.65	H–1 → L			
2^1B_{2u}	3.76	H → L+4			
2^1E_{1u}	3.79 (0.012)	H → L+5			
3^1E_{1u}	3.92 (0.201)	H → L+4			
2^1A_{1g}	4.07	H–1 → L			
3^1E_{2g}	4.21	H–2 → L, H → L+3			

a) Oscillator strength is given in parentheses for optically allowed states. b) TDDFT results (Ref. 23). c) Semiempirical molecular orbital calculations (Ref. 22). d) HOMO. e) LUMO. f) TD BOP calculation. g) TD LC-BOP calculation. h) MO/8E calculation. i) ZINDO calculation.

The SAC-CI results agree well with the experimental peak positions. We could clearly assign the α- and p-bands to the S_1 (1^1B_{2u}) and S_2 (1^1B_{1u}) states, respectively. In the D_{6h} structure, both the S_1 and S_2 states were optically forbidden. For the D_{3d} structure, the S_2 state was optically allowed. This symmetry lowering explained the relatively high intensity of the p-band observed in the experiment. We could also exclude the possibility of assigning the p-band to the S_3 state because a transition to the S_3 state is optically forbidden, even in the D_{3d} structure. Then, the β-band (β^0-band) could be assigned to the S_4 state (1^1E_{1u}). The previous study did not conclude whether the β-band corresponds to S_3 or S_4.[33] The present SAC-CI results agreed with the time-dependent density functional theory (TDDFT) and Zerner's intermediate neglect of differential overlap (ZINDO) calculations reported previously,[33] and therefore we have clarified the assignment for the β-band.

The β^1- and β^2-bands were assigned to vibronic excitations of β^0-band with one and two quanta of an a_{1g} mode.[33] The present calculation shows that 2^1E_{1u} and 3^1E_{1u} states exist around this energy region. This result suggest another possibility that the β^1- and β^2-bands may be due to another electronic state, the optically allowed 2^1E_{1u} and 3^1E_{1u} states, whose calculated intensities are much lower than that for the 1^1E_{1u} state. It is more consistent to assign the β^1- and β^2-bands to vibronic progressions of β^0-band because their high intensities may be attributable to the Franck–Condon factors.[44] The energy differences between the bands are about 0.14–0.19 eV (1130–1530 cm^{-1}). These energies are also consistent with the CC stretching modes of HBC.[45] The absorption of the 2^1E_{1u} and 3^1E_{1u} states would be covered by these vibronic bands and are not observed. Within the present results, we did not change the previous assignment for the β^1- and β^2-bands. More accurate computations with improved conditions and also detailed analysis of excited-state vibrations are necessary to provide a reliable assignment for these bands. The present calculations suggest that at least three electronic states exist around the β-band region. The appearance of the 2^1E_{1u} and 3^1E_{1u} states near β-band region was characteristic of HBC; the corresponding states for coronene and *cir*-coronene had a much higher energy than the β-band state.

Table 4 shows the SAC-CI results of *cir*-coronene. Experimental data that can be directly compared with our calculation was not available; therefore, the SAC-CI results were compared with semiempirical MO and TDDFT results.[22,23] The lowest three states, S_1, S_2, and S_3, had the character of the HOMO to LUMO transition. By analogy with the coronene case, the 1^1B_{2u}, 1^1B_{1u}, and 1^1E_{1u} states were termed, the α-, p-, and β-bands, respectively. A strong absorption of the β-band was calculated in the visible light region. The SAC-CI calculations showed a bathochromic shift (red shift) of the absorptions and an increase in intensity of optically allowed states with extension of the π-system.

TDDFT calculations have been reported for the lowest excited states. The importance of the long-range correction for the exchange term of DFT was noted.[23] The present SAC-CI result was rather close to the TDDFT without long-range correction. The SAC-CI excitation energies were lower than those of the semiempirical MO calculations about 0.1–0.3 eV for α- and β-states.[22] In contrast to coronene, the excitation energy for the α-state of *cir*-coronene seems to be underestimated. The reason for this underestimation is yet unclear.

Frozen Orbital Analysis for HOMO–LUMO Transitions. The low-lying excited states of these molecules are characterized as the electron transition between the doubly degenerate HOMO and doubly degenerate LUMO. Such a transition generates four electronic states. It is important to predict which state has the lowest excitation energy for spectroscopic and practical applications. Thus, the mechanism of their energy splitting was studied using FZOA, CIS, and SAC-CI. The FZOA takes into account the singly excited states within the degenerate orbital. In the present case, the CIS calculation with only two occupied and two unoccupied orbitals corresponds to the FZOA. That represents purely orbital interactions, without orbital-relaxation effects arising from the electronic excitation.[28,29] The excitation energies in FZOA are decomposed into three terms: $\Delta E = A + B + C$. Here, the A term is the orbital energy difference. The B term consists of J_{ai} and K_{ai}, where J and K denote the Coulomb and exchange integrals, respectively, between occupied and unoccupied orbitals. The C term contains off-diagonal contributions. These terms for the present systems, transitions between e_{2u} and e_{1g} orbitals, are summarized in Table 5. For FZOA of the present systems, the degenerate orbitals are classified in D_{2h} symmetry, which is an abelian subgroup of D_{6h} as follows:

$$(\mathrm{e_{1g}}) \rightarrow \begin{cases} \mathrm{b_{3g}} \equiv a \\ \mathrm{b_{2g}} \equiv b \end{cases} \tag{1}$$

and

$$(\mathrm{e_{1u}}) \rightarrow \begin{cases} \mathrm{b_{1u}} \equiv i \\ \mathrm{a_{u}} \equiv j \end{cases} \tag{2}$$

The values of the A, B, and C terms of FZOA for coronene, HBC, and *cir*-coronene are given in Table 6, and their schematic behaviors for the energy splitting of coronene and HBC are illustrated in Figure 2.

First, we examine the energy splitting for coronene. The contribution for the excitation energy from the A term was 8.62 eV. The B term split the energies of the $^1B_{1u}$ and $^1B_{2u}$ states, and also the doubly degenerate $^1E_{1u}$ state, by 0.08 eV. The C term reduced the excitation energy of the $^1B_{1u}$ and $^1B_{2u}$ states and raised the energy of the $^1E_{1u}$ states. The C term enhanced the splitting between $^1B_{1u}$ and $^1B_{2u}$ by 0.08 eV and removed the splitting of the $^1E_{1u}$ state. The contributions from the term B and C for the splitting between the $^1B_{1u}$ and $^1B_{2u}$ states must be the same magnitude, otherwise the degeneracy of the $^1E_{1u}$ state could not be recovered by the $B + C$ terms.

Table 5. FZOA Energy Components of HOMO–LUMO Transition of Coronene, HBC, and *Cir*-coronene

State	A	B	C
$^1B_{1u}$	$\varepsilon_a - \varepsilon_i$	$2K_{ia} - J_{ia}$	$-[2(jb\|ai) - (ab\|ij)]$
$^1B_{2u}$	$\varepsilon_a - \varepsilon_i$	$2K_{ja} - J_{ja}$	$-[2(ib\|aj) - (ab\|ij)]$
$^1E_{1u}$	$\varepsilon_a - \varepsilon_i$	$2K_{ia} - J_{ia}$	$2(jb\|ai) - (ab\|ij)$
$^1E_{1u}{}'$	$\varepsilon_a - \varepsilon_i$	$2K_{ja} - J_{ja}$	$2(ib\|aj) - (ab\|ij)$

The behaviors of the A, B, and C terms were basically common to coronene, HBC, and *cir*-coronene. However, the relative importance of these terms was different. The magnitudes of the A and B terms decreased with extension of the π-system. This trend could be explained by the HOMO–LUMO gaps and delocalization of the π-electrons. The magnitudes of the C term did not simply correlate with the molecular size. The values of the C term were smallest in HBC. Molecular shape will also affect the contribution of the C term, and therefore, the excitation energies depend on the molecular size and shape.

The excitation energies from the CIS calculations are shown in Table 6. The CIS calculations presented here considered the same active MOs as the SAC-CI calculations. Interestingly, the lowest state of coronene predicted by the CIS calculation is different from the SAC-CI and FZOA calculations. We may consider that the difference between the FZOA and

Table 6. FZOA Energy Component, CIS, and SAC-CI Excitation Energy in eV

State	FZOA				CIS	SAC-CI
	A	B	C	Total		
Coronene						
$^1B_{1u}$		−2.96	−0.87	4.77	4.16	3.87
$^1E_{1u}$	8.62		0.87	6.52	5.82	4.63
$^1E_{1u}{}'$		−3.04	0.95	6.52	5.82	4.63
$^1B_{2u}$			−0.95	4.62	4.19	3.12
HBC						
$^1B_{1u}$		−2.84	−0.54	4.63	4.03	3.26
$^1E_{1u}$	8.01		0.54	5.70	5.05	3.78
$^1E_{1u}{}'$		−2.88	0.57	5.70	5.05	3.78
$^1B_{2u}$			−0.57	4.56	3.82	2.72
Cir-coronene						
$^1B_{1u}$		−2.18	−0.64	3.75	3.20	2.43
$^1E_{1u}$	6.57		0.64	5.03	4.48	3.01
$^1E_{1u}{}'$		−2.23	0.68	5.03	4.48	3.01
$^1B_{2u}$			−0.68	3.67	3.21	1.91

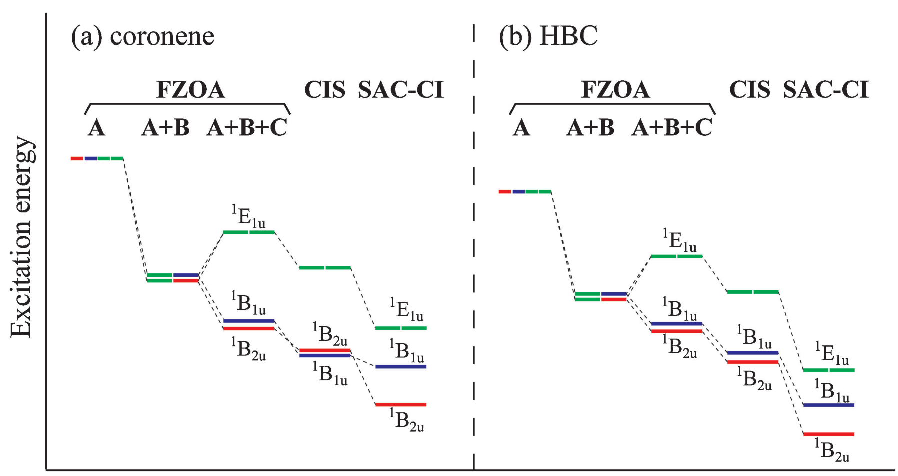

Figure 2. Illustration of the energy splitting of $^1B_{1u}$, $^1B_{2u}$, and $^1E_{1u}$ states with FZOA model, CIS, and SAC-CI calculations in (a) coronene and (b) HBC. The energy differences between the $^1B_{1u}$ and $^1B_{2u}$ states were magnified, and therefore they are not proportional to the actual values.

CIS represents an orbital-relaxation effect and the difference between the CIS and SAC-CI arises from an electron-correlation effect. The HOMO–LUMO interaction, considered by FZOA, made $^1B_{2u}$ the lowest state. In $^1B_{1u}$, the orbital relaxation, considered by the CIS, was larger than that in $^1B_{2u}$. Thus, the energy level was altered in the CIS results. The electron correlation in $^1B_{2u}$ was much larger than that in $^1B_{1u}$. Consequently, the energy level was again altered in the SAC-CI calculation. The electron correlation was the most significant factor in determining the lowest states, as can be seen in Figure 2.

Similar effects of orbital relaxation and electron correlation were observed for *cir*-coronene; an alternation in the energy level with CIS calculation occurred. For HBC, however, such energy-level alternation did not arise. The orbital relaxation in $^1B_{2u}$ was larger than that in $^1B_{1u}$ for HBC. Therefore, the splitting between $^1B_{1u}$ and $^1B_{2u}$ calculated by FZOA was enhanced by the CIS calculation. For HBC, both the electron correlation and orbital relaxation were significant in determining the lowest states.

Conclusion

The absorption spectra of coronene, HBC, and *cir*-coronene were studied by an ab initio quantum chemical method based on a cluster expansion theory of the SAC and SAC-CI methods. The SAC-CI calculations reasonably explained the experimental spectra and also predicted optically forbidden excited states. The principal findings of this study were summarized as follows:

(1) For coronene, the SAC-CI method calculated the vertical excitation energies that would correspond to the absorption maxima of the α- and *p*-bands with reasonable accuracy. The calculation overestimated the energy of the β-band by about 0.5 eV. Besides the limitation of the SAC-CI method, the result suggests that the solvent effect on this state could be large.

(2) For HBC, the symmetry lowering enhanced the intensity of the S_2 (1^1B_{2u}) state. This supported the assignment of the *p*-band to the S_2 state, rather than the S_3 state. The β-band could be assigned to the S_4 state. The present calculation showed that the optically allowed 2^1E_{1u} and 3^1E_{1u} states exist around the β-band region. The corresponding states for coronene and *cir*-coronene had a much higher energy than the β-band state.

(3) Significant bathochromic shifts of photoabsorptions and an increase in intensities of optically allowed states were predicted for *cir*-coronene by the SAC-CI calculation. It was predicted that *cir*-coronene has a strong absorption for the β-band in the visible light region.

(4) The mechanisms of energy splitting for the HOMO–LUMO transition were studied with the FZOA, CIS, and SAC-CI methods. The mechanisms depend on the molecular size and also molecular shape. The electron correlation considered by the SAC-CI was the most important origin for the energy splitting between the lowest and the next-lowest states. The CIS calculation could not correctly predict the relative energies of these states in coronene and *cir*-coronene. For HBC, the CIS provided the same energy order as the SAC-CI calculation.

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References

1 R. Rieger, K. Müllen, *J. Phys. Org. Chem.* **2010**, *23*, 315.
2 A. K. Geim, K. S. Novoselov, *Nat. Mater.* **2007**, *6*, 183.
3 J. C. Meyer, A. K. Geim, M. I. Katsnelson, K. S. Novoselov, T. J. Booth, S. Roth, *Nature* **2007**, *446*, 60.
4 J. R. Williams, L. DiCarlo, C. M. Marcus, *Science* **2007**, *317*, 638.
5 A. C. Grimsdale, K. Müllen, *Angew. Chem., Int. Ed.* **2005**, *44*, 5592.
6 J. Wu, W. Pisula, K. Müllen, *Chem. Rev.* **2007**, *107*, 718.
7 P. Ehrenfreund, S. B. Charnley, *Annu. Rev. Astron. Astrophys.* **2000**, *38*, 427.
8 B. T. Draine, *Annu. Rev. Astron. Astrophys.* **2003**, *41*, 241.
9 J. W. Patterson, *J. Am. Chem. Soc.* **1942**, *64*, 1485.
10 J.-i. Aihara, K. Ohno, H. Inokuchi, *Bull. Chem. Soc. Jpn.* **1970**, *43*, 2435.
11 K. Ohno, T. Kajiwara, H. Inokuchi, *Bull. Chem. Soc. Jpn.* **1972**, *45*, 996.
12 G. Bermudez, I. Y. Chan, *J. Phys. Chem.* **1986**, *90*, 5029.
13 G. Orlandi, F. Zerbetto, *Chem. Phys.* **1988**, *123*, 175.
14 E. Clar, *The Aromatic Sextet*, Wiley, New York, **1972**.
15 L. Schmidt-Mende, A. Fechtenkötter, K. Müllen, E. Moons, R. H. Friend, J. D. MacKenzie, *Science* **2001**, *293*, 1119.
16 J. P. Hill, W. Jin, A. Kosaka, T. Fukushima, H. Ichihara, T. Shimomura, K. Ito, T. Hashizume, N. Ishii, T. Aida, *Science* **2004**, *304*, 1481.
17 H. Seyler, B. Purushothaman, D. J. Jones, A. B. Holmes, W. W. H. Wong, *Pure Appl. Chem.* **2012**, *84*, 1047.
18 A. Keller, R. Kovacs, K.-H. Homann, *Phys. Chem. Chem. Phys.* **2000**, *2*, 1667.
19 J. R. Dias, *THEOCHEM* **2002**, *581*, 59.
20 A. Soncini, E. Steiner, P. W. Fowler, R. W. A. Havenith, L. W. Jenneskens, *Chem.—Eur. J.* **2003**, *9*, 2974.
21 F. Negri, E. di Donato, M. Tommasini, C. Castiglioni, G. Zerbi, K. Müllen, *J. Chem. Phys.* **2004**, *120*, 11889.
22 B. Hajgató, K. Ohno, *Chem. Phys. Lett.* **2004**, *385*, 512; X. Yang, B. Hajgató, M. Yamada, K. Ohno, *Chem. Lett.* **2005**, *34*, 506.
23 S. Tokura, T. Sato, T. Tsuneda, T. Nakajima, K. Hirao, *J. Comput. Chem.* **2008**, *29*, 1187.
24 R. Fukuda, M. Ehara, H. Nakatsuji, *J. Chem. Phys.* **2010**, *133*, 144316.
25 R. Fukuda, M. Ehara, *J. Chem. Phys.* **2012**, *136*, 114304.
26 R. Fukuda, M. Ehara, *J. Chem. Phys.* **2012**, *137*, 134304.
27 R. Fukuda, H. Nakatsuji, *J. Chem. Phys.* **2008**, *128*, 094105.
28 H. Nakai, H. Morita, H. Nakatsuji, *J. Phys. Chem.* **1996**, *100*, 15753.
29 T. Baba, Y. Imamura, M. Okamoto, H. Nakai, *Chem. Lett.* **2008**, *37*, 322.
30 Y. Imamura, T. Baba, H. Nakai, *Chem. Lett.* **2009**, *38*, 528.
31 C. Lee, W. Yang, R. G. Parr, *Phys. Rev. B* **1988**, *37*, 785; A. D. Becke, *J. Chem. Phys.* **1993**, *98*, 5648.

32 R. Ditchfield, W. J. Hehre, J. A. Pople, *J. Chem. Phys.* **1971**, *54*, 724; W. J. Hehre, R. Ditchfield, J. A. Pople, *J. Chem. Phys.* **1972**, *56*, 2257; P. C. Hariharan, J. A. Pople, *Theor. Chim. Acta* **1973**, *28*, 213; M. M. Francl, W. J. Pietro, W. J. Hehre, J. S. Binkley, M. S. Gordon, D. J. DeFrees, J. A. Pople, *J. Chem. Phys.* **1982**, *77*, 3654.

33 G. Rouillé, M. Steglich, F. Huisken, T. Henning, K. Müllen, *J. Chem. Phys.* **2009**, *131*, 204311.

34 H. Nakatsuji, K. Hirao, *Chem. Phys. Lett.* **1977**, *47*, 569; H. Nakatsuji, K. Hirao, *J. Chem. Phys.* **1978**, *68*, 2053.

35 H. Nakatsuji, *Chem. Phys. Lett.* **1978**, *59*, 362; H. Nakatsuji, *Chem. Phys. Lett.* **1979**, *67*, 329; H. Nakatsuji, *Chem. Phys. Lett.* **1979**, *67*, 334.

36 T. H. Dunning, Jr., P. J. Hay, in *Modern Theoretical Chemistry*, ed. by H. F. Schaefer, III, Plenum, New York, **1977**, Vol. 3, pp. 1–27.

37 H. Nakatsuji, *Chem. Phys.* **1983**, *75*, 425; H. Nakatsuji, J.-y. Hasegawa, M. Hada, *J. Chem. Phys.* **1996**, *104*, 2321.

38 M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H. P. Hratchian, A. F. Izmaylov, J. Bloino, G. Zheng, J. L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J. A. Montgomery, Jr., J. E. Peralta, F. Ogliaro, M. Bearpark, J. J. Heyd, E. Brothers, K. N. Kudin, V. N. Staroverov, R. Kobayashi, J. Normand, K. Raghavachari, A. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, N. Rega, J. M. Millam, M. Klene, J. E. Knox, J. B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R. E. Stratmann, O. Yazyev, A. J. Austin, R. Cammi, C. Pomelli, J. W. Ochterski, R. L. Martin, K. Morokuma, V. G. Zakrzewski, G. A. Voth, P. Salvador, J. J. Dannenberg, S. Dapprich, A. D. Daniels, Ö. Farkas, J. B. Foresman, J. V. Ortiz, J. Cioslowski, D. J. Fox, *Gaussian 09 (Development Version)*, Gaussian, Inc., Wallingford CT, **2012**.

39 A. Schäfer, C. Huber, R. Ahlrichs, *J. Chem. Phys.* **1994**, *100*, 5829.

40 K. Ohno, H. Inokuchi, *Chem. Phys. Lett.* **1973**, *23*, 561.

41 K. Kowalski, R. M. Olson, S. Krishnamoorthy, V. Tipparaju, E. Aprà, *J. Chem. Theory Comput.* **2011**, *7*, 2200.

42 Y. Ooshika, *J. Phys. Soc. Jpn.* **1954**, *9*, 594; E. G. McRae, *J. Phys. Chem.* **1957**, *61*, 562; S. Basu, *Adv. Quantum Chem.* **1964**, *1*, 145; T. Renger, B. Grundkötter, M. E.-A. Madjet, F. Müh, *Proc. Natl. Acad. Sci. U.S.A.* **2008**, *105*, 13235.

43 R. Fukuda, M. Ehara, *J. Chem. Theory Comput.* **2013**, *9*, 470.

44 K. Ohno, *Chem. Phys. Lett.* **1978**, *53*, 571; K. Ohno, *Chem. Phys. Lett.* **1979**, *64*, 560.

45 D. M. Hudgins, S. A. Sandford, *J. Phys. Chem. A* **1998**, *102*, 344.